`Problem 8` `Problems T4_PhaseDiagrams I.pdf #8` `Diagram construction`

Drawing a Eutectic Diagram and the 40 % B Alloy

`Original statement · Problems T4_PhaseDiagrams I.pdf, p. 3`

8. Two metals (A and B) have partial solubility among each other. Metal A, with a melting point at 700 °C, can dissolve 5% of B at 0 °C, and its maximum solubility occurs at 400 °C, being 25% B. On the other hand, B metal melts at 600 °C, can dissolve 10% of A at 0 °C and its maximum solubility occurs at 400 °C, being 20% A. At 400 °C there is an eutectic reaction when the B content is 65%. a. Draw and identify the parts of the phase diagram in equilibrium (suppose straight lines). b. For an alloy 40% B, determine the phases composition and their relative proportions at 450 °C, 400 °C + ΔT and 400 °C – ΔT. c. For an alloy 40% B, draw its microstructure indicating the microconstituents and the mass fraction of each one. Additionally, indicate the mass fraction of each of the phases present in such microconstituents.

Official Solution: b) 450 °C: C_L **= 55%B;** C_α **= 20%B;** $X_\alpha = 0.43$**;** $X_L = 0.57$**; 400 °C + ΔT:** C_L **= 65%B;** C_α **= 25%B;** $X_\alpha = 0.62$**;** $X_L = 0.38$**; 400 °C - ΔT:** C_β **= 80%B;** C_α **= 25%B;** $X_\alpha = 0.73$ **;** $X_\beta = 0.27$**. c) Eutectic microconstituent and proeutectic α;** $X_{\alpha p} = 0.62$**;** $X_E = 0.38$**; Phases in the eutectic microconstituent:** $X(\alpha_E) = 0.11$**;** $X(\beta_E) = 0.27$**.**

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Binary eutectic system** with partial solid solubility: α (rich in A) and β (rich in B) [Session 7 Slide 26].
2. **Straight lines** between the given points, as the statement asks.
3. Compositions in % B (mass); "20 % A" in β means 80 % B, and "10 % A" means 90 % B.

Points of the diagram

Point	Composition (% B)	Temperature (°C)	Meaning
Melting point of A	0	700	top of α solidus and of the A-side liquidus
Melting point of B	100	600	top of β solidus and of the B-side liquidus
Eutectic point E	65	400	lowest point of the liquid
Max. solubility of B in α	25	400	end of the eutectic line, α side

Point	Composition (% B)	Temperature (°C)	Meaning
Max. solubility of A in β	80	400	end of the eutectic line, β side
α solvus at 0 °C	5	0	
β solvus at 0 °C	90	0	

PHASE 2 · FRAMES & STRATEGY

Frame: composition (% B) on the horizontal axis and temperature (°C) on the vertical one. Each boundary is a straight segment, so a composition on it at a temperature T comes from linear interpolation between its two end points; the tie lines are horizontal. No change of basis is needed.

Lines of the diagram (part a):

- liquidus: $(0, 700) \to (65, 400)$ and $(100, 600) \to (65, 400)$;
- solidus: $(0, 700) \to (25, 400)$ for α and $(100, 600) \to (80, 400)$ for β;
- eutectic line: $(25, 400) \to (80, 400)$;
- solvus: $(25, 400) \to (5, 0)$ for α and $(80, 400) \to (90, 0)$ for β.

Regions: L above the liquidus; α + L and β + L between liquidus and solidus; α and β next to the axes, bounded by solidus and solvus; α + β below the eutectic line, between the solvus lines.

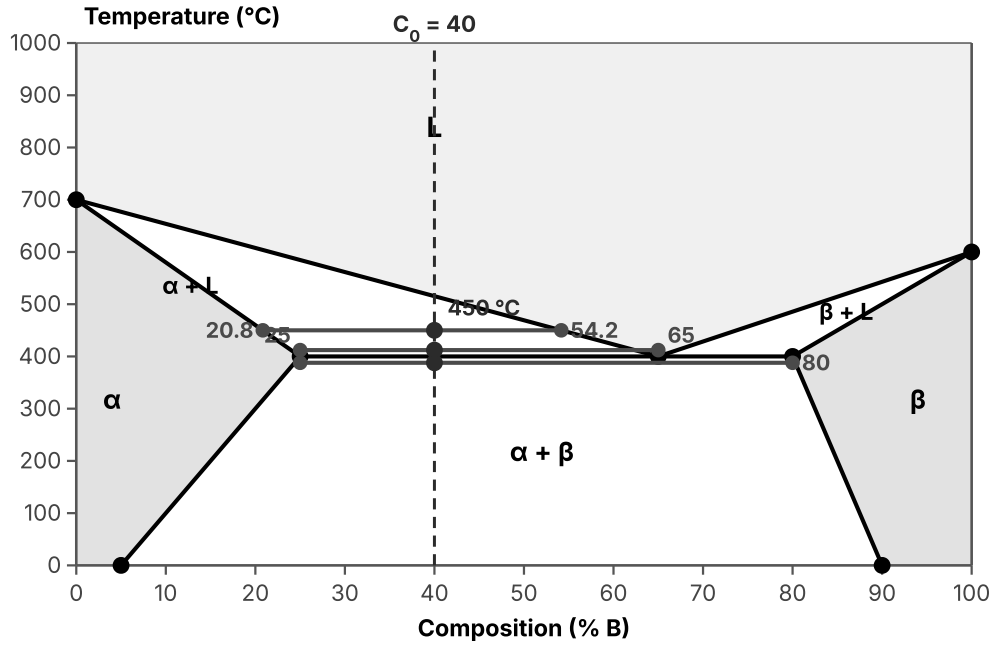


Figure 13. A-B diagram with straight lines: melting points 700 °C (A) and 600 °C (B), eutectic at 400 °C and 65 % B, maximum solubilities 25 % B in α and 20 % A in β (80 % B) at 400 °C, and 5 % B in α and 10 % A in β (90 % B) at 0 °C. Dashed: the 40 % B alloy, with its tie lines at 450 °C and just above and below 400 °C.

PHASE 3 · STEP-BY-STEP DERIVATION

b) Phases of the 40 % B alloy 450 °C. The alloy is between the A-side liquidus and the α solidus: **α + L**. On a straight segment from $(0, 700)$ to $(C_1, 400)$ the composition at T is $C = C_1 \dfrac{700 - T}{700 - 400}$:

$$C_L = 65 \times \frac{700 - 450}{300} = 65 \times \frac{250}{300} = 54.17\ \%\text{B} \qquad C_\alpha = 25 \times \frac{250}{300} = 20.83\ \%\text{B}$$

$$W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{40 - 20.83}{54.17 - 20.83} = \frac{19.17}{33.33} = 0.575 \qquad W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{54.17 - 40}{33.33} = \frac{14.17}{33.33} = 0.425$$

400 °C + ΔT. Still α + L, with the ends of the tie line at the eutectic temperature: $C_\alpha = 25$, $C_L = 65$ % B (length 40):

$$W_L = \frac{40-25}{65-25} = \frac{15}{40} = 0.375 \qquad W_\alpha = \frac{65-40}{40} = \frac{25}{40} = 0.625$$

400 °C – ΔT. α + β; $C_\alpha = 25$, $C_\beta = 80$ % B (length 55):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{80-40}{80-25} = \frac{40}{55} = 0.727 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{40-25}{55} = \frac{15}{55} = 0.273$$

c) Microconstituents at 400 °C – ΔT $25 < 40 < 65$: hypoeutectic, so the microconstituents are **proeutectic α** (25 % B) and **eutectic** (65 % B overall). The α present just above 400 °C is the proeutectic α, and the liquid becomes eutectic [Session 7 Slide 32]:

$$W_{\alpha p} = \frac{C_e - C_0}{C_e - C_\alpha} = \frac{65-40}{65-25} = 0.625 \qquad W_E = \frac{C_0 - C_\alpha}{C_e - C_\alpha} = \frac{40-25}{40} = 0.375$$

Phases inside the eutectic:

$$W_{\alpha,E} = W_\alpha - W_{\alpha p} = 0.7273 - 0.6250 = 0.1023 \qquad W_{\beta,E} = W_\beta = 0.273$$

The same numbers from the make-up of the eutectic, which is $(80-65)/(80-25) = 15/55 = 0.273$ α and $40/55 = 0.727$ β: $W_{\alpha,E} = 0.375 \times 0.273 = 0.102$ and $W_{\beta,E} = 0.375 \times 0.727 = 0.273$.

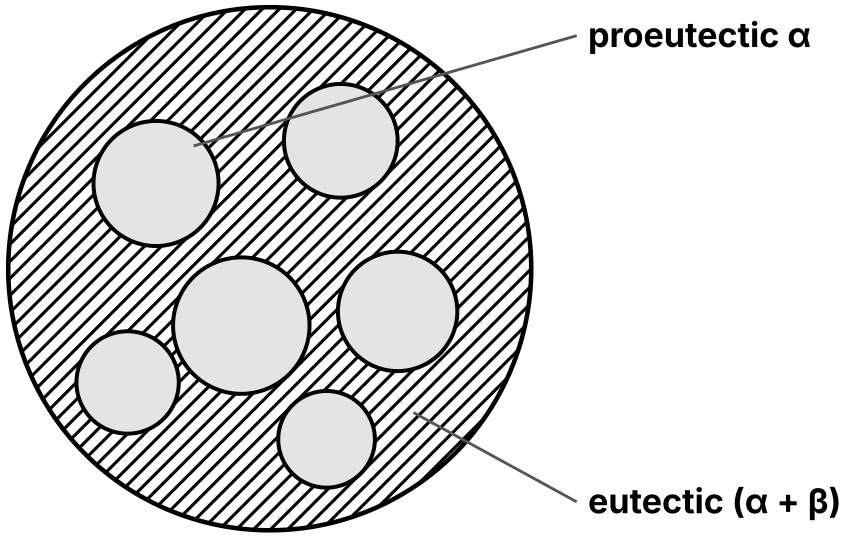


Figure 14. 40 % B alloy at 400 − ΔT: proeutectic α (62.5 %) in the eutectic α + β (37.5 %).

PHASE 4 · CHECKS & INTERPRETATION

Checks

- Mass balance at 450 °C: $0.425 \times 20.83 + 0.575 \times 54.17 = 8.85 + 31.15 = 40.0$ % B.
- Mass balance at 400 °C − ΔT: $0.727 \times 25 + 0.273 \times 80 = 18.18 + 21.82 = 40.0$ % B.
- $W_{\alpha,E} + W_{\beta,E} = 0.102 + 0.273 = 0.375 = W_E$, and $W_{\alpha p} + W_{\alpha,E} = 0.625 + 0.102 = 0.727 = W_\alpha$.
- **Limiting cases** (fractions dimensionless, compositions in % B): as C_0 approaches either end of a tie line, the phase at that end tends to the whole alloy and the other to 0; the 40 % B alloy lies inside every tie line used, so all its fractions are between 0 and 1.

Interpretation On cooling from the liquid, α starts to form at the liquidus ($700 - 300 \times 40/65 = 515$ °C for this alloy), grows to 62.5 % at the eutectic temperature, and the remaining 37.5 % of liquid, of 65 % B, becomes eutectic at 400 °C. Most of the eutectic is β (0.273 of the alloy), which is all the β of the alloy.

Check against the official solution

At 400 °C ± ΔT the results agree with the key (0.62/0.38 are 0.625/0.375; 0.73/0.27). The key writes "C_α = 25%Cu" at 400 °C + ΔT, a typo for 25 % B.

DISCREPANCY WITH THE OFFICIAL SOLUTION

450 °C. The key gives $X_L = 0.57$, $X_\alpha = 0.43$, which correspond to $C_L = 55$ and $C_\alpha = 20$ % B, presumably read graphically from its drawing (Solution problems Phase diagrams I.pdf, pp. 14-25, show no 450 °C tie line). With the straight lines that the statement requires, linear interpolation gives $C_L = 54.17$ and $C_\alpha = 20.83$ % B, hence $W_L = 0.575$ and $W_\alpha = 0.425$. The difference comes from reading the drawing; the interpolated values are kept.

α in the eutectic. The key gives $X(\alpha_E) = 0.11$, which is $0.73 - 0.62$ with rounded values. Exactly, $0.7273 - 0.6250 = 0.1023$, confirmed by $0.375 \times 15/55 = 0.102$; the key's 0.11 is a rounding effect.